Turtle Tunnel: helping hatchlings reach the ocean

A nest box and collapsible tunnel that protect sea turtle nests and guide hatchlings to the water.

The problem

Sea turtles are endangered, and they are a keystone species. Their ecosystems depend on them. Local rehabilitation centers send volunteers to watch over nests. Volunteers mark each site and clear a path to the beach so hatchlings have the best chance of reaching the ocean.

Hatchlings still face threats on the way to the water.

Threats hatchlings face

- Light pollution disorients hatchlings
- Predators threaten eggs and hatchlings
- People: nest disturbance, holes, sandcastles, chairs, and trash

Our solution

The system has two parts:

- **Nest box** — wood frame and mesh sides (airflow, less heat). A half wood panel on every side reduces light leaking in and stops hatchlings from crawling out the back or sides. Volunteers can clip on a full solid panel with magnets on whichever side faces artificial light. A hinged mesh lid opens for nest access. The box sits over the nest, raised on wood posts.
- **Collapsible tunnel** — runs from the box to the waterline. It telescopes to adjust length with the tide. Loops and stakes anchor it to the beach. Caution-color paint makes it visible to people. Small guiding lights along the inside ceiling replace the ocean glow the tunnel would otherwise block. Two floor options: open bottom so hatchlings crawl on real sand (held by stakes), or a full-circle tunnel with sand added inside, level with the beach.

Side view and cross-sections. Numbers match the table below. Full labeled drawing with two tunnel-floor options: *hatchling_path_design.png*.

Design details

#	Design challenge	How we address it
1	High water can block the tunnel mouth	Collapsible tunnel adjusts length with the tide
2	A light tunnel can be moved by wind	Loops and stakes hold it down; sand added during expansion adds weight
3	Hatchlings need to crawl on sand	Option A: open bottom so they crawl on real beach. Option B: full-circle tunnel with sand added inside, level with beach
4	The tunnel blocks the natural light hatchlings use to find the water	Guiding lights along the inside ceiling replace the ocean glow
5	A box could keep volunteers from reaching the nest	Hinged mesh lid opens for access
6	A box can get hot and let in artificial light	Mesh sides for airflow; half wood panel on every side. Full solid panels (magnets) on sides facing artificial light
7	Hatchlings could crawl out the back or sides of the box	Half wood panel on every side; solid wood wall (no mesh) on tunnel side
8	A predator could enter through the tunnel mouth	Box protects the nest; risk remains at the mouth — asking the rehab center
9	The tunnel and box can be tripping hazards	Caution-color paint and caution stripe on tunnel top

We want your feedback. We are asking Turtle Beach Rehab (Mary and Kathy) to review this design. Our questions: Would this help volunteers? Could the box harm eggs (temperature, airflow, moisture)? Could the guiding lights harm or confuse hatchlings — what color and brightness is safe, or should there be no lights? Open bottom or sand floor? What mesh size keeps predators out but lets hatchlings escape? Should the box be buried or staked to stop digging predators? Is wood safe near nests? Could the box or tunnel trap or entangle a hatchling or nesting mother?